

SPSS 3 T-Test

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Statistical Decision

- Make decisions about populations based on the sample information.
- eg. test whether the exam result of a class is higher than the passing mark.
- In addition to the answer of yes or no, how confident is the answer?

Statistical Decision

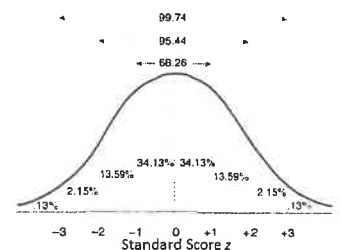
- Assumptions:
 - The population(s) is in normal distribution.
 - The mean and standard deviation of the population are known.
- Question to be answered:
 - Now, we have a sample value. Does the sample belong to the population?

Statistical Decision

- Formulation of hypothesis from the question:
 - An educated guess/assumption on the results to be verified.
 - Preferably be formulated before carrying out the experiment.
 - For a test on means, as an example, there are two statistical hypothesis:
 - Null hypothesis H_0 : the two means are equal ($\bar{X} = \bar{Y}$)
 - Alternative hypothesis H_1 : the two means are different (either $\bar{X} > \bar{Y}$, $\bar{X} < \bar{Y}$ or $\bar{X} \neq \bar{Y}$)

Statistical Decision

- Significant level (SL)
 - The probability of rejecting a hypothesis while it is correct (type I error), or *vice versa* (type II error)
 - eg. if we reject a hypothesis at 0.05 SL, there is 5% chance that the hypothesis is correct.



Statistical Decision

- Procedure for test of means
 - With defined SL, look up the range of standard scores (critical values) $\pm z_{SL}$ from the standard table.
 - Convert the sample value x into standard score z .

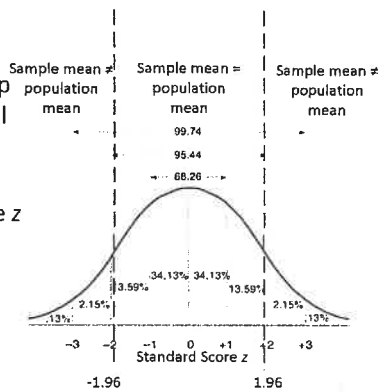
$$z = \frac{x - \bar{X}}{\sigma_X}$$
 where σ_X is the standard deviation of the population
 - If the standard score of the sample is within the range, it is statistically equal to the population mean $\bar{X}$.

Fig. 2. 5

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Statistical Decision

- **Example**
 - With SL of 0.05, look up the table for the critical values: ± 1.96 (2-tailed)
 - Convert the sample value to standard score z
 - Check whether the sample score z falls between the critical values
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Idea of T-Test

- Difficult to obtain population mean and SD.
 - Number of sample N is small.
 - Estimate the population parameters from the samples and use Student's *t* distribution.
- Eg. Population mean $\bar{X}$ = Sample mean;
 SD of means $\sigma_{\bar{X}}$ = Sample SD $/\sqrt{N}$
- *t* distribution is similar to normal distribution, with compensation for sample size using degree of freedom (df).
 - Usage of *t* score is similar to *z* score

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Pseudonym

Type of T-Test

- One sample T-test
- Paired samples T-Test
- Independent samples T-Test

34-2205

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One Sample T-Test

- Given the limited information on a population and a test value, T-Test can tell if the value differs from the population or not, and the variability of the difference
- Assume the population is in normal distribution
- eg. test whether the scores of students is greater than the preset goal of 60% average

11/28/2017

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One Sample T-Test

- Required information
 - The population
 - A sample value
 - Confidence interval percentage (100% - significant level)



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Example Data File

- Data file: truancy.sav
- It shows the percentage of truant days of students before and after intervention by the parents
- Variables:
 - grade: the study level of the student
 - gender: gender of the student
 - prepct: percent truant days before intervention
 - postpct: percent truant days after intervention

Task: One Sample T-Test

- Perform one sample T-Test at 95% confidence interval (0.05 significant level) on a sample of value 13
- 1. File > Open: "truancy.sav"
- 2. Analyze > Compare Means > One-sample T-Test
- 3. Set Test Variable to prepct, set Test Value to 13
- 4. Options > set Confidence Interval Percentage to 95% > continue > OK

Results of One Sample T-Test

- Sample statistics table
 - Ordinary sample stats: N, mean, std. deviation
 - Std. Error Mean: estimated standard deviation of mean of sample group (Std. Deviation/ $\sqrt{N}$)

One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
prepct	289	14.2038	13.97168	.75593

Sample stats

Std. Error of Mean

Result of One Sample T-Test

- One-sample Test
 - df: degree of freedom (N-1)
 - Mean Difference: the difference of the test value and the mean (14.2038-13)
 - t: T-score of the mean difference, similar to standard score
(mean difference / std. error of mean)

One-Sample Test						
Test Value = 13						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
prepct	1.592	288	.112	1.20378	-.2829	2.6915

Result of One Sample T-Test

- One-sample test (Con't)
 - Sig. (2-tailed): the 2-tailed significant level, indicating the probability, that the test value is the same as the population but we think they are not the same (wrongly rejecting the null hypothesis)
 - 95% Confidence Interval of the Difference: 95% of the real mean differences of the population falls between the limits
 - Sig. >0.05 Accept the null hypothesis (no difference in means) (statistically same)

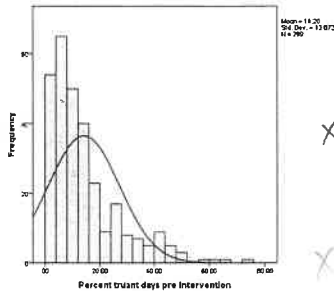
One-Sample Test						
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prepct	1.592	288	.112	1.20378	-.2829	2.6915

Task: Assumption of One Sample T-Test

- Check normal distribution of the population
 1. Graphs > Legacy Dialogs > Histogram
 2. Set Variable to prepct
 3. Check Display Normal Curve > ok

Assumption of One Sample T-Test

- Check normal distribution of the population (Con't)

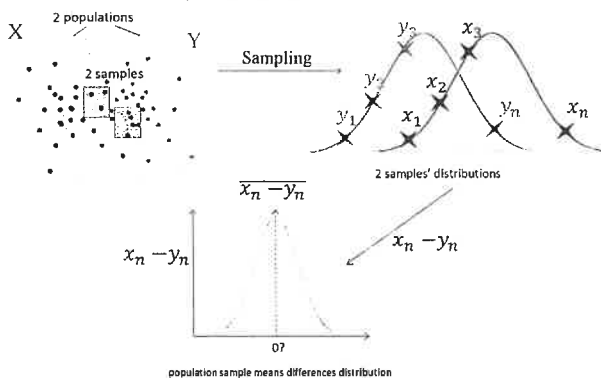


X normal distribution
X analyze

Paired Sample T-Test

- There are two experimental conditions and SAME participants take part in both conditions.
- The means and variability of the two samples groups are compared. "S.D."
- Assume the differences, but not the sample values, are in normal distribution
- eg. test whether the scores of students improved from the first test to the second test.

Are the 2 populations different?



Task: Paired Sample T-Test

- Perform paired sample T-Test at 95% confidence interval (0.05 significant level) between before and after intervention
- Analyze > Compare Means > Paired-samples T Test**
- Set **Paired Variables** to pretpct for variable 1, and postpct for variable 2
- Options > set Confidence Interval Percentage to 95% > continue > OK**

Results of Paired Sample T-Test

- Sample statistics table
 - Similar to the one-sample test
 - Repeated for both populations, every pair

Paired Samples Statistics				
	Mean	N	Std. Deviation	Std. Error Mean
Pair 1 pretpct	14.2039	299	13.07160	.72595
postpct	11.4378	299	11.18297	.66173

- Paired samples correlations
 - Correlation of the two populations (Pearson's r)
 - Be caution if the sig. level is too large (>0.05), indicating the poor correlation.

Paired Samples Correlations			
	N	Correlation	Sig.
Pair 1 pretpct & postpct	299	.461	.008

Results of Paired Sample T-Test

- Paired Sample Test
 - Most of the parameters are similar to the ones of one-sample test
 - Std. Deviation: the standard deviation of the difference
 - With such a large t score (3.768) and hence low sig. level, the null hypothesis is rejected – the 2 populations are statistically different.

Paired Samples Test									
		Paired Differences							
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
					Lower	Upper			
Pair 1	pretpct - postpct	2.76602	12.65355	.73499	1.32137	4.21057	3.768	298	.008

meaningful to do the analysis

Find whether the correlation is strong/weak